

BIMBINGAN TUGAS AKHIR #4

1 Marginal Likelihood (Type II Maximum Likelihood / ML-II)

(Rasmussen & Williams, 2006, Bab 5.4.1, hlm. 112–113, Persamaan (5.7)–(5.8))

$$p(\mathbf{y} \mid X, \boldsymbol{\theta}) = \int p(\mathbf{y} \mid \mathbf{f}) p(\mathbf{f} \mid X, \boldsymbol{\theta}) d\mathbf{f} \quad (1)$$

$$\mathbf{y} \mid X, \boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, K_y), \quad K_y = K + \sigma_n^2 I_N \quad (2)$$

$$\log p(\mathbf{y} \mid X, \boldsymbol{\theta}) = -\frac{1}{2} \mathbf{y}^T K_y^{-1} \mathbf{y} - \frac{1}{2} \log |K_y| - \frac{N}{2} \log(2\pi) \quad (3)$$

$$\text{NLML}(\boldsymbol{\theta}) = \frac{1}{2} \mathbf{y}^T K_y^{-1} \mathbf{y} + \frac{1}{2} \log |K_y| + \frac{N}{2} \log(2\pi) \quad (4)$$

2 Maximum A Posteriori (MAP)

(Rasmussen & Williams, 2006, Bab 5.2, hlm. 108–109, Persamaan (5.2))

$$p(\boldsymbol{\theta} \mid \mathbf{y}, X) = \frac{p(\mathbf{y} \mid X, \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathbf{y} \mid X)} \quad (5)$$

$$\mathcal{L}_{\text{MAP}}(\boldsymbol{\theta}) = \log p(\mathbf{y} \mid X, \boldsymbol{\theta}) + \log p(\boldsymbol{\theta}) \quad (6)$$

$$\boldsymbol{\theta}_{\text{MAP}} = \arg \max_{\boldsymbol{\theta}} \left[-\frac{1}{2} \mathbf{y}^T K_y^{-1} \mathbf{y} - \frac{1}{2} \log |K_y| + \log p(\boldsymbol{\theta}) \right] \quad (7)$$

3 Leave-One-Out Cross-Validation: Squared Error

(Rasmussen & Williams, 2006, Bab 5.3 & Bab 5.4.2, hlm. 111 & hlm. 116–117, Persamaan (5.5) & (5.12))

$$\mu_{-i} = y_i - \frac{[K_y^{-1} \mathbf{y}]_i}{[K_y^{-1}]_{ii}} \quad (8)$$

$$L_{\text{LOO-SE}}(\boldsymbol{\theta}) = \sum_{i=1}^N (y_i - \mu_{-i})^2 = \sum_{i=1}^N \left(\frac{[K_y^{-1} \mathbf{y}]_i}{[K_y^{-1}]_{ii}} \right)^2 \quad (9)$$

4 LOO Cross-Validation: Negative Log Pseudo-Likelihood

(Rasmussen & Williams, 2006, Bab 5.3 & Bab 5.4.2, hlm. 111 & hlm. 116–118, Persamaan (5.6), (5.11) & (5.13))

$$\sigma_{-i}^2 = \frac{1}{[K_y^{-1}]_{ii}} \quad (10)$$

$$\begin{aligned}
L_{\text{LOO-NLPL}}(\boldsymbol{\theta}) &= - \sum_{i=1}^N \log p(y_i \mid X, \mathbf{y}_{-i}, \boldsymbol{\theta}) \\
&= \frac{1}{2} \sum_{i=1}^N \left(\log(2\pi\sigma_{-i}^2) + \frac{(y_i - \mu_{-i})^2}{\sigma_{-i}^2} \right) \\
&= \frac{1}{2} \sum_{i=1}^N \left(-\log[K_y^{-1}]_{ii} + \frac{([K_y^{-1}\mathbf{y}]_i)^2}{[K_y^{-1}]_{ii}} + \log(2\pi) \right) \tag{11}
\end{aligned}$$

Pustaka

- [1] Rasmussen, C. E., & Williams, C. K. (2006). *Gaussian Processes for Machine Learning*. MIT Press, Cambridge, MA.